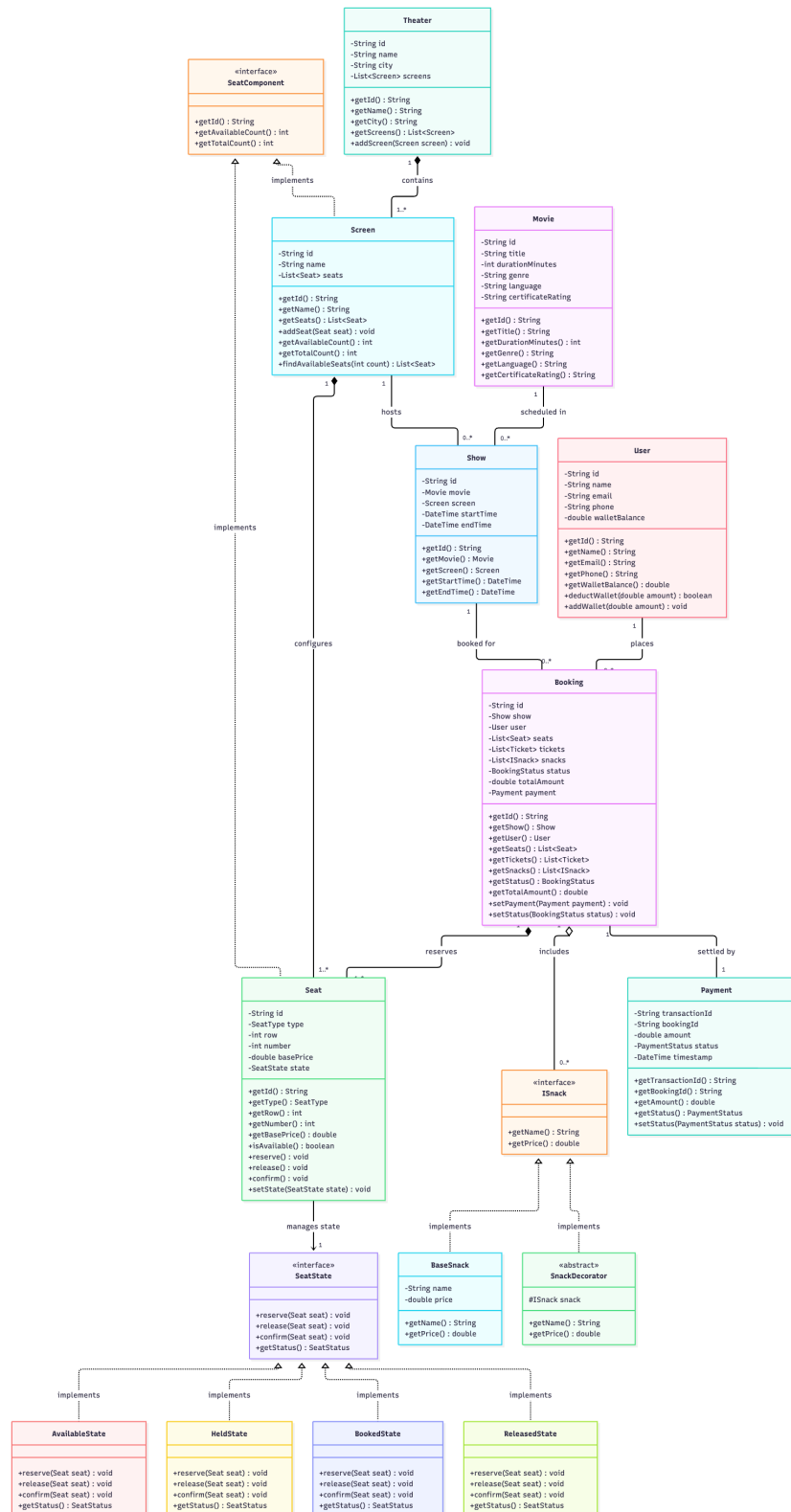


LLD Part 1: UML Class Diagram & Core Domain Model

Core Entities: Movie, Theater, Screen, Seat (with Row), Show, User, Booking, Payment, Snack.

Key Associations: Theater has-many Screens (Composition); Screen has-many Seats (Composite Pattern); Seat delegates to SeatState (State Pattern); Booking aggregates Tickets, Seats, Snacks, and Payment.



LLD Part 2: Unified Booking Sequence & Pattern Flow

Trace: Seat Selection → Composite Layout Check → Temporary Seat Hold (State) → Snack Customization (Decorator) → Sequential Payment Processing (Chain of Responsibility) → Polymorphic Ticket Issuance (Factory) → Booking Assembly (Builder) → Decoupled Event Dispatch (Observer).

Rollback: On payment rejection or hold timeout, BookingInvoker triggers reverse compensation releasing held seats.

